

CSCE 5300: Introduction to Big Data and Data Science

Chapter 5 (5.1): Data Learning Methods



Qutline



- **5.1.1. Bayes rule**
- **5.1.5. Naïve Bayes**

5.1.1. Bayes Rule



- **Theorem 5.1.** (Bayes Rule). Denote by X and Y random variables, then the following holds

$$p(y|x) = \frac{p(x|y)p(y)}{p(x)}. \quad (5.1.1)$$

- This follows from the fact that $p(x, y) = p(x|y)p(y) = p(y|x)p(x)$, where $p(y|x)$, means "the probability of event y given that event x has already happened."
- The key consequence of (5.1.1) is that we may reverse the conditioning between a pair of random variables [1] – [3].

- [1] Shai Shalev-Shwartz and Shai Ben-David, *Understanding Machine learning: From theory to Algorithms*, Cambridge University Press, New York, 2014.
- [2] Alex Smola and S.V.N. Vishwanathan, *Introduction to Machine learning*, Cambridge University Press, Cambridge, UK, 2010.
- [3] Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach*: 4th Edition, Prentice Hall, 2020.

5.1.1. Bayes Rule



- **Example 5.1**
- We illustrate our reasoning by means of a simple example – conclusion using an AIDS (acquired immunodeficiency syndrome) test.
- Assume that a patient would like to have such a test carried out on him.
- The physician recommends a test which is guaranteed to detect HIV (human immunodeficiency virus infection)-positive whenever a patient is infected.
- On the other hand, for healthy patients it has a 1% error rate.
- That is, with probability 0.01 it diagnoses a patient as HIV-positive even when he is, in fact, HIV-negative.
- Moreover, assume that 0.15% of the population is infected.

5.1.1. Bayes Rule



- Now assume that the patient has the test carried out and the test returns 'HIV - negative'.
 - Denote by X and T the random variables associated with the health status of the patient and the outcome of the test respectively.
 - We are interested in $p(X = \text{HIV+} | T = \text{HIV+})$.
 - By Bayes rule we may write
- $$p(X = \text{HIV+} | T = \text{HIV+}) = \frac{p(T = \text{HIV+} | X = \text{HIV+})p(X = \text{HIV+})}{p(T = \text{HIV+})} \quad (5.1.2)$$

5.1.1. Bayes Rule



- While we know all terms in the numerator, $p(T = \text{HIV}+)$ itself is unknown.
- That said, it can be computed via

$$\begin{aligned} p(T = \text{HIV}+) &= \sum_{x \in \{\text{HIV}+, \text{HIV}-\}} p(T = \text{HIV}+, x) \\ &= \sum_{x \in \{\text{HIV}+, \text{HIV}-\}} p(T = \text{HIV}+|x)p(x) \\ &= 1.0 \cdot 0.0015 + 0.01 \cdot 0.9985. \end{aligned} \tag{5.1.3}$$

- Substituting back into the conditional expression yields

$$p(X = \text{HIV}+|T = \text{HIV}+) = \frac{1.0 \cdot 0.0015}{1.0 \cdot 0.0015 + 0.01 \cdot 0.9985} = 0.1306. \tag{5.1.4}$$

5.1.1. Bayes Rule



- In other words, even though our test is quite reliable, there is such a low prior probability of having been infected with AIDS that there is not much evidence to accept the hypothesis even after this test.
- Let us now think how we could improve the diagnosis.
- One way is to obtain further information about the patient and to use this in the diagnosis.
- For instance, information about his age is quite useful.
- Suppose the patient is 35 years old.
- In this case we would want to compute $p(X = \text{HIV+IT} = \text{HIV+}, A = 35)$ where the random variable A denotes the age.
- The corresponding expression yields:

5.1.1. Bayes Rule



- $$\frac{p(T = \text{HIV+} | X = \text{HIV+}, A)p(X = \text{HIV+} | A)}{p(T = \text{HIV+} | A)} \quad (5.1.5)$$

- Here we simply conditioned all random variables on A in order to take additional information into account.
- We may assume that the test is independent of the age of the patient, i.e.

- $$p(t|x, a) = p(t|x). \quad (5.1.6)$$

5.1.1. Bayes Rule



- What remains therefore is $p(X = \text{HIV+}|A)$.
- US census data pegs this number at approximately 0.9%.
- Plugging all data back into the conditional expression yields $1 \times 0.009 / (1 \times 0.009 + 0.01 \times 0.991) = 0.48$.
- By including additional observed random variables our estimate becomes more reliable.

5.1.1. Bayes Rule



- A second tool in our arsenal is the use of multiple measurements.
- After the first test the physician is likely to carry out a second test to confirm the diagnosis.
- We denote by $T1$ and $T2$ (and $t1$; $t2$ respectively) the two tests.
- Obviously, what we want is that $T2$ will give us an “independent” second estimate.
- We want to ensure that $T2$ does not make the same mistakes as $T1$.
- It is probably not good idea to repeat $T1$ without changes, since it might perform the same diagnostic mistake as before.
- The diagnosis of $T2$ should be independent of that of $T1$ given the health status X of the patient.

5.1.1. Bayes Rule

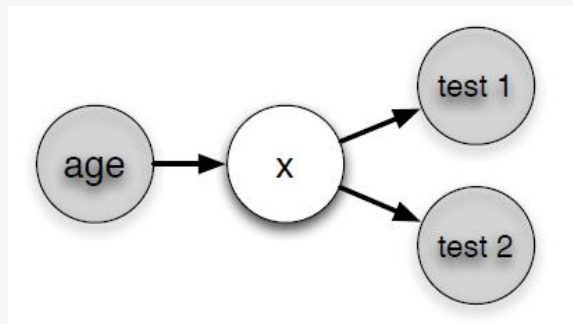


- This is expressed as

- $$p(t_1, t_2 | x) = p(t_1 | x)p(t_2 | x). \quad (5.1.7)$$

- A graphical illustration of the setting is shown in Fig. 5.1.1.
- Random variables satisfying the condition (5.1.7) are commonly referred to as conditionally independent.
- For the argument we assume that the statistics for T2 is presented in Fig. 5.1.5.

5.1.1. Bayes Rule



- Fig. 5.1.1. A graphical description of our HIV testing scenario. Knowing the age of the patient influences our prior on whether the patient is HIV positive (the random variable X). The outcomes of the tests 1 and 2 are independent of each other given the status X . We observe the shaded random variables (age, test 1, test 2) and would like to estimate the un-shaded random variable X .

5.1.1. Bayes Rule



$p(t_2 x)$	$x = \text{HIV-}$	$x = \text{HIV+}$
$t_2 = \text{HIV-}$	0.95	0.01
$t_2 = \text{HIV+}$	0.05	0.99

- Fig. 5.1.5. Statistics for T5.

5.1.1. Bayes Rule



- This test is less reliable than the first one.
- However, we may now combine both estimates to obtain a very reliable estimate based on the combination of both events.
- For instance, for $t_1 = t_2 = \text{HIV+}$ we have

- $$p(X = \text{HIV+} | T_1 = \text{HIV+}, T_2 = \text{HIV+}) = \frac{1.0 \cdot 0.99 \cdot 0.009}{1.0 \cdot 0.99 \cdot 0.009 + 0.01 \cdot 0.05 \cdot 0.991} = 0.95. \quad (5.1.8)$$

5.1.1. Bayes Rule



- By combining two tests we can now confirm with very high confidence that the patient is indeed diseased.
- What we have carried out is a combination of evidence.
- Strong experimental evidence of two positive tests effectively overcame an initially very strong prior which suggested that the patient might be healthy.

5.1.5. Naïve Bayes



- ***Gaussian Naïve Bayes***
- It is the simplest Naïve Bayes classifier having the assumption that the data from each label is drawn from a simple Gaussian distribution.
- In this case, the Bayes classifier can be formulated by stating the following:
- ***If the output $P(x/y)$ is larger than given threshold T , assign the variable vector y to class $L1$. Otherwise, assign it to class $L5$.***

5.1.5. Naïve Bayes



- ***Pros***
- The followings are some pros of using Naïve Bayes classifiers –
- Naïve Bayes classification is easy to implement and fast.
- It will converge faster than discriminative models.
- It requires less training data.
- It is highly scalable in nature, or they scale linearly with the number of predictors and data points.
- It can make probabilistic predictions and can handle continuous as well as discrete data.
- Naïve Bayes classification algorithm can be used for binary as well as multi-class classification problems.

5.1.5. Naïve Bayes



- **Cons**
- The followings are some cons of using Naïve Bayes classifiers –
- One of the most important cons of Naïve Bayes classification is its strong feature independence because in real life it is almost impossible to have a set of features which are completely independent of each other.
- Another issue with Naïve Bayes classification is its ‘zero frequency’ which means that if a categorical variable has a category but not being observed in training data set, then Naïve Bayes model will assign a zero probability to it and it will be unable to make a prediction.

5.1.5. Naïve Bayes



- ***Applications of Naïve Bayes classification***
- The following are some common applications of Naïve Bayes classification –
- **Real-time prediction** – Due to its ease of implementation and fast computation, it can be used to do prediction in real-time.
- **Multi-class prediction** – Naïve Bayes classification algorithm can be used to predict posterior probability of multiple classes of target variable.
- **Text classification** – Due to the feature of multi-class prediction, Naïve Bayes classification algorithms are well suited for text classification. That is why it is also used to solve problems like spam-filtering and sentiment analysis.
- **Recommendation system** – Along with the algorithms like collaborative filtering, Naïve Bayes makes a Recommendation system which can be used to filter unseen information and to predict whether a user would like the given resource or not.

5.1.5. Naïve Bayes

- Fig. 5.1.3.
- Naïve Bayes algorithm
- to solve document search problem

```
Train( $\mathbf{X}, \mathbf{Y}$ ) {reads documents  $\mathbf{X}$  and labels  $\mathbf{Y}$ }
  Compute dictionary  $D$  of  $\mathbf{X}$  with  $n$  words.
  Compute  $m, m_{\text{ham}}$  and  $m_{\text{spam}}$ .
  Initialize  $b := \log c + \log m_{\text{ham}} - \log m_{\text{spam}}$  to offset the rejection threshold
  Initialize  $p \in \mathbb{R}^{2 \times n}$  with  $p_{ij} = 1, w_{\text{spam}} = n, w_{\text{ham}} = n$ .
  {Count occurrence of each word}
  {Here  $x_i^j$  denotes the number of times word  $j$  occurs in document  $x_i$ }
  for  $i = 1$  to  $m$  do
    if  $y_i = \text{spam}$  then
      for  $j = 1$  to  $n$  do
         $p_{0,j} \leftarrow p_{0,j} + x_i^j$ 
         $w_{\text{spam}} \leftarrow w_{\text{spam}} + x_i^j$ 
      end for
    else
      for  $j = 1$  to  $n$  do
         $p_{1,j} \leftarrow p_{1,j} + x_i^j$ 
         $w_{\text{ham}} \leftarrow w_{\text{ham}} + x_i^j$ 
      end for
    end if
  end for
  {Normalize counts to yield word probabilities}
  for  $j = 1$  to  $n$  do
     $p_{0,j} \leftarrow p_{0,j} / w_{\text{spam}}$ 
     $p_{1,j} \leftarrow p_{1,j} / w_{\text{ham}}$ 
  end for
Classify( $x$ ) {classifies document  $x$ }
  Initialize score threshold  $t = -b$ 
  for  $j = 1$  to  $n$  do
     $t \leftarrow t + x^j (\log p_{0,j} - \log p_{1,j})$ 
  end for
  if  $t > 0$  return spam else return ham
```

5.1.5. Naïve Bayes



- ***Building model using Naïve Bayes in Python***
- Python library, Scikit learn can be used to build a Naïve Bayes model in Python.
- The python code and output data can be presented as follows:
- *CSCE 5300 – Naive Bayes*
- # load the iris dataset
- from sklearn.datasets import load_iris
- iris = load_iris()
- # store the feature matrix (X) and response vector (y)
- X = iris.data
- y = iris.target

5.1.5. Naïve Bayes



- # splitting X and y into training and testing sets
- from sklearn.model_selection import train_test_split
- X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.4, random_state=1)
-
- # training the model on training set
- from sklearn.naive_bayes import GaussianNB
- gnb = GaussianNB()
- gnb.fit(X_train, y_train)
-
- # making predictions on the testing set
- y_pred = gnb.predict(X_test)

5.1.5. Naïve Bayes



- `# comparing actual response values (y_test) with predicted response values (y_pred)`
- `from sklearn import metrics`
- `print("Gaussian Naive Bayes model accuracy(in %):", metrics.accuracy_score(y_test, y_pred)*100)`
- Gaussian Naive Bayes model accuracy(in %): 95.5.0